

STUDY-UNI · FULL MARKING SCHEME

FE3020

Understanding Supply Chain Management

Institution	University College Cork
Examination session	Winter 2022
Paper number	Paper 1
Duration of paper	1.5 hours
Special requirements	Calculator required
Instructions to candidates	Answer two questions. All questions carry equal marks.
Questions covered in this scheme	Question 1 (part (a) plus BOTH alternative routes, (b) and (c)), Question 2, Question 3

This is an original marking scheme prepared for exam-preparation purposes. It is not an official UCC document and has no connection to the module's examiners or external examiner.

Question 1

60% + (40% from EITHER (b) OR (c)) = 100%

SOURCING NOTE Question 1 requires part (a) AND either part (b) OR part (c) — not both. This scheme marks (a) at 60% and then marks (b) and (c) separately, each at 40%, so it is usable whichever route a candidate takes; only one of (b)/(c) should be added to (a) when totalling an actual script.

(a)**60%**

Identify the main stages in the evolution of operations planning and for each of these indicate, (i) the relevant decade(s), (ii) operations and planning control focus, (iii) Information System design type, and (iv) key concepts and approaches. You should use a summary table as part of your answer.

FULL MODEL ANSWER

Framework (module content, after Olhager 2013). Operations planning and control has evolved through five broad stages, each widening in scope — from controlling stock of individual items, to controlling the end product, to eventually planning across the whole supply chain.

Decade(s)	Operations & planning control focus	Information System design type	Key concepts & approaches
1960s	Shop floor control of independent demand items	Inventory control	Statistical inventory control, EOQ (Economic Order Quantity), ROP (Re-Order Point), safety stocks, Gantt charts, sequencing/scheduling rules
1970s	MRP — end product, focus via the Bill of Materials	MRP (Materials Requirement Planning)	Closed-loop MRP, BOM explosion, dynamic lot sizing, CRP (Capacity Requirements Planning)
1980s	Master Production Scheduling — end product, external focus on the customer	MRP II (Manufacturing Resource Planning)	ATP (Available to Promise), JIT/TPS (Toyota Production System), OPT/TOC (Theory of Constraints)
1990s	S&OP — product-group level	ERP (Enterprise Resource Planning)	Lean, agile, leagile
2000s onward	Supply chain business — all product levels	ERP, SCP (Supply Chain Planning), e-business	CPFR, VMI

Narrative. The progression is cumulative rather than a series of replacements: each stage widens the planning scope of the one before it. The 1960s focus is purely on controlling stock of individual, independent-demand items. MRP in the 1970s introduces the Bill of Materials as the organising structure, so planning is driven by what the end product actually requires rather than by each component's stock level in isolation. MRP II in the 1980s extends this from materials alone to the full manufacturing resource picture, and explicitly brings in the customer-facing side (Available to Promise). S&OP and ERP in the 1990s lift planning to product-group level and integrate it with the wider business system. From the 2000s, the unit of planning becomes the supply chain itself — spanning all product levels and multiple firms

— supported by collaborative approaches like CPFR and VMI rather than planning tools internal to a single company.

Mark Allocation — Part (a) (worth 60% of the question)

Criterion	Marks	What earns the marks
Correct stages/decades identified	15%	All five stages (1960s–2000s+) named with correct decades
Operations & planning control focus per stage	12%	Shop floor/independent items → BOM/end product → MPS/customer → S&OP/product group → supply chain/all levels
Information System design type per stage	12%	Inventory control → MRP → MRP II → ERP → ERP/SCP/e-business, correctly matched to each decade
Key concepts & approaches per stage	13%	At least one correct, decade-appropriate concept per stage (EOQ/ROP, closed-loop MRP, JIT/ATP, Lean/agile, CPFR/VMI)
Summary table used, and narrative shows the progression is cumulative	8%	Table present AND narrative explains the widening scope across stages, not just a list
Total	60%	Sum of criteria above

EITHER PART (b) OR PART (c)

(b)**40%**

Given three demand profiles (runner, repeater, stranger) for products supplied, identify and justify the appropriate manufacturing strategy that you would recommend for each of these for a Continuous Replenishment supply chain.

FULL MODEL ANSWER

Context. A Continuous Replenishment supply chain is characterised by short lead times and relatively predictable overall demand at the network level — but individual SKUs within that network differ hugely in volume and predictability, which is exactly what the runner/repeater/stranger classification captures.

Demand profile	Demand characteristics	Recommended manufacturing strategy	Justification
Runner	High volume, predictable demand	Manufacture To Stock (MTS)	Forecast accuracy is high enough, and volume large enough, that producing ahead of firm orders and holding short-term finished-goods stock minimises lead time to the customer without carrying material forecast risk — the classic case for building to a forecast.
Repeater	High or low volume, but follows a recognisable demand pattern	MTO or MTS depending on capability; postponement/ Configure To Order (CTO) where possible	Repeaters sit between the two extremes — regular enough to plan against, but not always high-volume enough to justify pure MTS. Where the manufacturing process allows late-stage differentiation, postponement lets a largely-generic item be finished to the specific order only once real demand is known, capturing MTS-level responsiveness without carrying finished-goods risk on every variant.
Stranger	Often low volume, unpredictable demand	Manufacture To Order (MTO) — hold manufacturing until the order is received, if possible	With low, unpredictable demand, a forecast-driven MTS approach carries a high risk of building stock that never sells (obsolescence, write-off). Waiting for a firm order removes that risk entirely, at the cost of a longer lead time to the customer — an acceptable trade-off given how infrequently and unpredictably strangers are actually ordered.

Why this matters specifically for a Continuous Replenishment chain. The whole premise of continuous replenishment — short lead times, high service levels, frequent small deliveries — depends on not treating every SKU the same way. Applying MTS uniformly across the full range (as the module's own product-portfolio content notes, ranges tend to grow, with new variants outpacing the rate of elimination) would build excess, increasingly obsolete stock on the long tail of low-volume strangers. Applying MTO uniformly would fail to hit the short lead times runners' customers expect. Segmenting by demand profile lets the manufacturing strategy match the risk profile of each SKU rather than forcing a single strategy to serve an increasingly long and varied product range.

Mark Allocation — Part (b) [if chosen] (worth 40% of the question)

Criterion	Marks	What earns the marks
Runner — correct strategy and justification	13%	MTS correctly recommended, justified by high-volume/predictable demand
Repeater — correct strategy and justification	13%	MTO/MTS with postponement (CTO) correctly recommended, justified by intermediate/pattern-following demand
Stranger — correct strategy and justification	10%	MTO correctly recommended, justified by low-volume/unpredictable demand and obsolescence risk
Links explicitly to the Continuous Replenishment context	4%	Explains why uniform strategy would fail this specific supply chain type, not just the three profiles in isolation
Total	40%	Sum of criteria above

(c)**40%**

A large-scale consumer food manufacturing site has initiated a review of their buying policy. Currently buyers are paid a bonus if they achieve a price discount on purchases of 10% or higher. The buyers have taken advantage of the scale of the manufacturing site to place large orders with suppliers, sometimes enough in one delivery to satisfy site demand for an entire year. The site manager has asked you to review this practice and introduce a purchasing and inventory management policy that reduces costs while maintaining material availability targets.

As a starting point, you consider the total cost of ownership of materials. What are the main components of total cost of ownership for direct goods? How would you use this concept when devising a policy proposal for the site manager?

FULL MODEL ANSWER

Components of total cost of ownership for direct goods. The module's framework breaks TCO for a direct good (raw material) into:

- **Purchase price** — the unit price agreed with the supplier.
- **Cost of acquisition** — the transactional cost of placing and receiving the order.
- **Carrying costs**, comprising: material handling, warehouse storage costs, obsolescence & shrinkage, insurance, and cost of capital.

What the current buying practice does to each component. The buyer bonus rewards purchase-price discount alone. Ordering a full year of site demand in one delivery does minimise that single line — but it does so by directly increasing every carrying-cost element: a year's stock must be handled and stored (material handling, warehouse storage), sits exposed to damage, spoilage and obsolescence for far longer than a shorter-cycle order would (obsolescence & shrinkage — a material concern for consumer *food* product given shelf-life constraints), carries a higher insured value (insurance), and ties up working capital for the full year rather than releasing it progressively (cost of capital). This is a textbook case of local optimisation: the incentive rewards minimising one TCO line at the direct expense of the others, and the buyers' behaviour is a rational response to that specific incentive, not a one-off error.

SOURCING NOTE *Framing the bonus scheme as a 'local optimisation' problem is standard purchasing/procurement reasoning applied to this scenario; it is not a named model in the module slides, and should be credited on the strength of the TCO-component reasoning, not as recall of a named framework.*

Using TCO to devise the policy proposal.

- **Redesign the bonus metric** away from "discount achieved" toward a net TCO saving measure, so that a smaller purchase-price discount achieved through a materially lower carrying-cost profile scores at least as well as today's 10%+ discount bonus.
- **Move from single annual bulk orders toward smaller, more frequent deliveries**, sized to reflect realistic shelf-life and demand-coverage horizons for a food manufacturing site rather than pure order-cost minimisation — directly reducing warehouse storage,

obsolescence/shrinkage, insurance, and cost-of-capital exposure, at the cost of a smaller purchase-price discount and marginally higher per-order acquisition cost.

- **Differentiate ordering policy by material risk:** perishable or shelf-life-constrained materials should carry the tightest ordering discipline (smallest order size, shortest coverage horizon), while genuinely long-shelf-life, stable materials can safely support larger orders where the TCO math still favours it.
- **Maintain material availability targets via TCO-informed safety stock** — driven by supplier lead time and criticality — rather than by bulk-buying, so that availability and low unit cost are pursued through separate, appropriate levers rather than conflated into a single discount-chasing behaviour.

Mark Allocation — Part (c) [if chosen] (worth 40% of the question)

Criterion	Marks	What earns the marks
TCO components for direct goods correctly listed	12%	Purchase price, cost of acquisition, and all five carrying-cost elements identified
Explains why current practice is a local optimisation problem	12%	Connects large one-off orders to increases in each of the non-price TCO components, including shelf-life/obsolescence relevance for food
TCO used to devise the policy proposal	12%	Concrete, TCO-anchored proposals (incentive redesign, order-frequency change, risk-differentiated ordering)
Addresses material availability target explicitly	4%	Shows the proposal still protects availability, not just cost
Total	40%	Sum of criteria above

Question 1 — total (a + one of b/c)

100%

Question 2

60% + 40% = 100%

(a)**60%**

Supply chain development has been characterised by specialisation and outsourcing. Using the example of grocery supply chains, illustrate and discuss the supply chain design before and after the introduction of central distribution, highlighting performance implications. Your answer should include basic before and after supply chain design schematics and identify key metric(s) when discussing performance.

FULL MODEL ANSWER

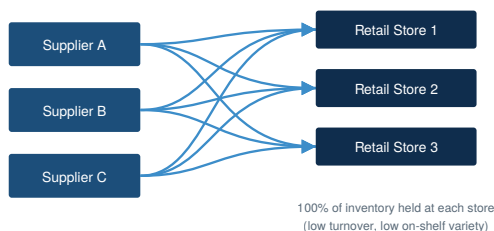
Specialisation and outsourcing framing. Central distribution is itself an example of the specialisation and outsourcing trend: rather than every supplier managing its own delivery relationship with every store, distribution and consolidation is concentrated into a specialist function (frequently an appointed 3PL), and each party specialises in the part of the chain it does best — suppliers in production, the 3PL in distribution, the retailer in store operations.

Before central distribution. Consumer food processors deliver directly to individual retail stores, typically limited to around twice-weekly delivery. Each store holds and manages its own inventory independently — there is no pooling of stock across the retailer's network, and delivery frequency is constrained by the economics of visiting many individual stores directly.

After central distribution. Delivery shifts to daily supplier deliveries into a central (or regional) distribution centre, with onward daily delivery to retail stores driven by daily store orders. Logistics arrangements range from delivery into the RDC to backhaul by the retailer's own distribution fleet from the supplier's factory gate (Factory Gate Pricing), and RDC operations frequently involve cross-docking — immediate despatch of supplier orders from the RDC to stores without being taken into stock — often run by an appointed 3PL.

BEFORE — Direct Store Delivery

Limited to c. 2 deliveries / week per store



AFTER — Central Distribution

Daily supplier → RDC delivery; daily RDC → store, order-driven

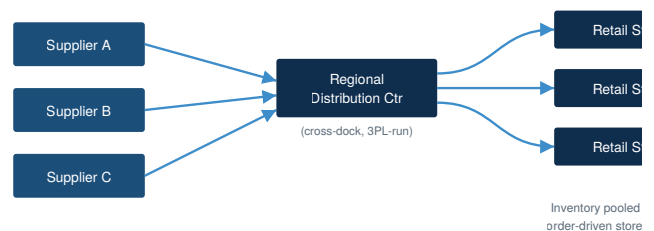


Figure 1 — Grocery supply chain design before and after the introduction of central distribution. Arrows indicate physical product flow; inventory pools at store level before, and at the RDC after.

Performance implications:

- **Reduced days inventory on-hand throughout the chain** — pooling stock at the RDC rather than fragmenting it across every individual store lowers total inventory required for the same service level, and cross-docking further reduces RDC dwell time.

- **Improved information flow, reducing bullwhip** — daily, order-driven replenishment gives much tighter, higher-frequency demand signals back up the chain than twice-weekly direct delivery, reducing the amplification and lag that drive the bullwhip effect.
- **Enhanced stock turnover and on-shelf variety at retail** — more frequent, smaller, order-driven deliveries let stores carry a wider range without a proportional increase in backroom stock.
- **Requires continuous replenishment capability and high supplier compliance** — daily, order-driven delivery only works if suppliers reliably deliver on time and in full; this is why OTIF becomes a key operating metric under this model.
- **Opportunity and dependency for suppliers** — closer, higher-frequency integration with a large retailer creates the basis for a category-leadership relationship, though it also concentrates dependency on that retailer's central distribution requirements — a direct consequence of the specialisation trend the question asks you to frame this within.

Key metrics. The module identifies inventory turnover, the cash-to-cash cycle time, and (retailer-specific) sales per m² as the core metrics for assessing the performance implications of this shift, alongside OTIF as the compliance metric that continuous replenishment depends on.

Mark Allocation — Part (a) (worth 60% of the question)

Criterion	Marks	What earns the marks
Frames the answer around specialisation/outsourcing	5%	Explicitly connects central distribution to the specialisation/outsourcing trend named in the question
Accurately describes the 'before' design	10%	Direct-to-store delivery, twice-weekly, store-level inventory correctly identified
Accurately describes the 'after' design	10%	Daily RDC-based delivery, cross-docking, 3PL/backhaul correctly identified
Schematic diagrams provided (before and after)	10%	Basic before/after schematics included and broadly correct in structure
Performance implications discussed	15%	At least three implications discussed with correct reasoning
Key metric(s) correctly identified	10%	Inventory turnover AND cash-to-cash cycle AND/OR sales per m ² AND OTIF, correctly tied to the discussion
Total	60%	Sum of criteria above

(b)**40%**

MH Eng, a fabrication manufacturer, is a niche player in the market. Their point of differentiation is the high level of customisation offered to their customers. This commands a premium price. Over a number of years, they have built strong business relationships with a range of specialist suppliers of metals and components. Many of these suppliers are small scale. This supplier base responds quickly to orders which enables MH Eng to use make-to-order fulfilment strategy.

The business was acquired last year by a large multinational corporation. Finance in HQ have introduced an accounts payable policy of 50 days. Currently MH Eng holds inventory on hand for 40 days on average, and its customers usually pay within 50 days. Offsetting these figures is an average payables period of 30 days.

Calculate the cash-to-cash cycle time. Given the context outlined above, how would you respond to the request for Finance at Corporate HQ?

FULL MODEL ANSWER

Cash-to-cash cycle formula (module content): Days Inventory On-hand + Days Sales Outstanding – Days Payables Outstanding = Cash-to-cash days.

Scenario	DIO	DSO	DPO	Cash-to-cash
Current (30-day supplier terms)	40	50	30	60 days
Proposed (Group 50-day policy)	40	50	50	40 days

$40 + 50 - 30 = 60$ **days** currently. $40 + 50 - 50 = 40$ **days** under the Group's proposed policy — a **20-day reduction** in the cash-to-cash cycle, which releases cash and is numerically favourable from a Group cash-conversion perspective.

Verdict — Numerically favourable to Finance, but risks the supplier network MH Eng's premium positioning depends on

How to respond to Corporate HQ. The calculation is straightforward, but the underlying trade-off is the same one that recurs across this module's supplier-finance scenarios: Finance's objective (conserve funds, improve the Group's reported cash position) is not automatically aligned with Supply Chain's objective (continuity and quality of supply at a fair price). MH Eng's competitive differentiation — high customisation, make-to-order fulfilment, premium pricing — depends specifically on small-scale specialist metal and component suppliers who are explicitly described as responding quickly to orders, which by construction means they carry little buffer stock or financial slack of their own. Extending their payment terms from 30 to 50 days is a direct cashflow shock to exactly the suppliers MH Eng's business model is least able to lose.

The appropriate response is a differentiated implementation, not a flat refusal or a silent roll-out:

- **Segment suppliers before applying the new terms** — apply the 50-day policy to larger, financially robust suppliers, while protecting the small-scale specialists whose responsiveness is the actual source of MH Eng's premium positioning.
- **Offer supply chain finance / early-payment facilities** for suppliers where 50-day terms would otherwise threaten continuity — the supplier is paid early via a financing arrangement while MH Eng still books the 50-day term for Group cash-to-cash purposes.
- **Make the integration risk explicit to HQ:** MH Eng was acquired specifically for its differentiated, premium proposition, which is itself built on this supplier network. A blanket Group-wide payables policy applied without reference to that network risks eroding the very asset the acquisition was made for, and this needs to be surfaced as a named risk to Corporate HQ rather than absorbed quietly at site level.

Mark Allocation — Part (b) (worth 40% of the question)

Criterion	Marks	What earns the marks
Correct formula applied	5%	DIO + DSO – DPO stated and applied correctly
Correct current cash-to-cash calculation	5%	$40 + 50 - 30 = 60$ days, shown with full working
Correct proposed cash-to-cash calculation	5%	$40 + 50 - 50 = 40$ days, shown with full working, and the 20-day change correctly identified
Response is specific to MH Eng's supplier base	15%	Explicitly connects the risk to small-scale specialist suppliers and the make-to-order/premium proposition, not a generic answer
Proposes a workable response to HQ	10%	At least one concrete mechanism (segmentation, supply chain finance, escalation of acquisition/integration risk)
Total	40%	Sum of criteria above

Question 2 — total

100%

Question 3

30% + 30% + 40% = 100%

(a)**30%**

Oxfam's Global Supermarket Scorecard is an example of third sector influence on food supply chain ethics. Identify the four criteria used by Oxfam to score food retailer ethical behaviour and comment on why these are/are not important food supply chain issues.

FULL MODEL ANSWER

The four criteria (module content). Oxfam's Supermarket Scorecard focuses on the social dimension of sustainable development, scoring retailers across four themes:

Criterion	What it assesses	Why it matters
Transparency & accountability	Whether retailers are open about their sourcing policies and practices	Retailers with considerable leverage over their supply chains can and should be transparent about where they source from, what they expect from suppliers, and whether their own policies actually enable — rather than hinder — suppliers' ability to meet those expectations. Without transparency, none of the other three criteria can be independently verified.
Workers	Labour conditions across global food supply chains	Tens of millions of people work in global food supply chains, and their conditions fall along a spectrum from forced labour at one end to decent, fairly remunerated work at the other. This criterion matters because retail purchasing practices (price pressure, order volatility) directly shape which end of that spectrum workers experience, even though the retailer rarely employs them directly.
Farmers (small-scale)	The viability of small-scale agriculture within retailer supply chains	The vast majority of the world's farms are small-scale, and farming is critical to the livelihoods of hundreds of millions of the world's poorest people. Supermarket purchasing practices affect whether small-scale producers remain viable, whether they can organise collectively, and whether they receive adequate government support and advocacy.
Women	Human and labour rights impacts specific to women in food supply chains	Women make up a large proportion of workers in the food and agriculture sector, yet gender-specific risks (pay gaps, discrimination, unsafe conditions) are easily hidden inside aggregate labour statistics. A dedicated criterion is needed because a supply chain can score adequately on "workers" in general while still failing women specifically.

Why third-sector scoring matters here. All four criteria concern conditions occurring far upstream of the retailer, in supplier and sub-supplier facilities the retailer does not directly operate — exactly the kind of issue that governance-only ESG disclosure (see this paper's ESG discussion elsewhere in the module) can struggle to verify. An independent NGO scorecard supplies external pressure and public visibility that a retailer's own governance reporting, on its own, would not.

Mark Allocation — Part (a) (worth 30% of the question)

Criterion	Marks	What earns the marks
Four criteria correctly named	10%	Transparency & accountability, Workers, Farmers, Women — all four correctly identified
Correct explanation of each criterion	12%	Each of the four correctly explained (3% each), not just listed
Comment on importance	8%	Explains why these are genuine food-supply-chain issues (leverage, upstream invisibility, livelihoods, gender-specific risk)
Total	30%	Sum of criteria above

(b)**30%**

The 'True Cost' concept, as popularised by Andrew Morgan (2015), has sparked considerable debate about 'fast fashion'. Identify two key supply chain sustainability issues that this concept highlights and how it seeks to address them.

FULL MODEL ANSWER

SOURCING NOTE Andrew Morgan's 2015 documentary *The True Cost* and its specific application to fast fashion are not named in the module slides, which present the True Cost/externalities framework primarily through food-system sources (the True Price Foundation and the Rockefeller Foundation's True Cost of Food work). This answer applies the module's own True Cost framework to the fast fashion context the question specifies; the documentary detail was checked independently and the framework content is the module's own.

The concept. True cost economics represents the gap between a product's market price and the total societal cost of producing it — including negative externalities such as environmental harm and impacts on public health that are not reflected in the price paid. *The True Cost* (2015), directed by Andrew Morgan, applied this lens to the fast fashion industry, tracing low-cost, rapidly-cycled clothing back to its garment-worker and environmental impacts, prompted in part by the 2013 Rana Plaza factory collapse in Bangladesh.

Two key supply chain sustainability issues:

- **Human rights / labour conditions (social externality).** Fast fashion's low retail prices are enabled in part by unsafe factory conditions and very low wages in garment-producing countries — a cost borne by workers rather than reflected in the price consumers pay. This is the module's "social" category of externality: underpayment, poor working conditions, and health and safety risk sitting outside the transaction price entirely.
- **Environmental damage (environmental externality).** The fast fashion production cycle — pesticide-intensive cotton farming, dye and chemical pollution of water and soil, and a linear take-make-waste model ending in landfill — imposes environmental costs on producing regions and future consumers that are likewise absent from the shelf price. This is the module's "environmental" externality category: pollution, resource overuse, and a linear rather than circular model.

How the True Cost approach seeks to address them. Rather than treating these as unpriced side-effects, a true cost assessment quantifies the gap between market price and true societal cost for a given garment, making the externality visible and therefore actionable. The module's framework proposes doing this by building preference (business and consumer) for lower true-price-gap products, enabling businesses to reduce their own true-price gaps directly, enabling remediation for affected workers or communities where harm has already occurred, and — where a business chooses to — allowing consumers to pay the true-price gap on top of the transaction price to fund that remediation directly. Applied to fast fashion specifically, this reframes "cheap clothing" as clothing whose true cost has simply been shifted onto garment workers and the environment rather than genuinely eliminated.

Mark Allocation — Part (b) (worth 30% of the question)

Criterion	Marks	What earns the marks
Two key sustainability issues correctly identified	12%	Human rights/labour AND environmental damage, each correctly linked to fast fashion specifically
Explains the True Cost concept correctly	8%	Market-price vs societal-cost gap, negative externalities, correctly defined
Explains how the approach seeks to address the issues	10%	At least two of the module's proposed mechanisms (preference-building, remediation, true-price-gap payment) applied to fast fashion
Total	30%	Sum of criteria above

(c)**40%**

The WE_RACE Company builds yachts and other pleasure craft. They are a well-established business and for many years supplied a set portfolio of design options to a range of customers. Last year they hired a new lead design engineer who has worked closely with the sales team. Working together they offered increasing levels of customisation, including craft with bespoke components and features. They pride themselves on quick design and delivery of this increasing variety of pleasure craft. This has boosted sales, but finance have reported a continual erosion of profit margin. Furthermore, the warehouse manager has rented additional space due to the range of components required. The Managing Director has asked Finance for key numbers so she can have a better sense of the current situation. They have given her the following:

In 2019, the total cost of goods sold was €40 million. Its inventory holding average was €3 million. At the end of 2020, sales increased and the cost of goods sold was €50 million, while average inventory was €5 million.

Using the formula for inventory turnover, calculate the inventory turnover for this scenario. Comment on the performance achieved, including identification of possible reasons for this and measures that might be taken to improve performance.

FULL MODEL ANSWER

Formula (module content): Inventory Turnover = Cost of Goods Sold ÷ Average Inventory Held. The higher the turnover, the lower the firm's inventory costs, for a given level of sales.

Year	COGS	Avg. inventory	Turnover	≈ Days inventory
2019	€40m	€3m	13.3×	27.4
2020	€50m	€5m	10.0×	36.5

2019: €40m ÷ €3m = **13.3×** (to 1 decimal place). 2020: €50m ÷ €5m = **10.0×**. Turnover fell from roughly 13.3× to 10.0×, equivalent to inventory rising from roughly 27.4 days on hand to roughly 36.5 days.

Verdict — Inventory turnover performance has declined, despite rising sales

Comment on the performance achieved. Cost of goods sold grew 25% year-on-year (€40m → €50m), consistent with the boosted sales the scenario describes. But average inventory grew faster still — 67% (€3m → €5m) — so the business is now carrying proportionally far more stock to support the same revenue base than it was the year before. This is a genuine performance decline, not simply a side-effect of growth: rising sales and falling inventory efficiency are two separate outcomes that have both occurred here, and the second is quietly eroding the value created by the first — directly consistent with the "continual erosion of profit margin" Finance has reported.

Possible reasons for this. The scenario supplies a clear driver: the shift toward increasing customisation, including bespoke components and features, introduced by the new lead

design engineer working with sales. A widening range of bespoke, low-commonality components increases the total inventory needed to support the same level of sales, since many of these components can only be used on a subset of craft and cannot be pooled or substituted the way a smaller, more standardised range could be. The additional warehouse space rented is the physical, costed evidence of this build-up — not a separate cause, but a direct consequence of the same underlying driver. This mirrors a pattern that recurs across this module's inventory-turnover content: rising customisation, pursued for its sales and margin upside, tends to quietly erode margin again through the carrying cost of the low-volume, bespoke inventory it requires.

Measures to improve performance.

- **Monetise the cost of the additional inventory** (holding cost, incremental warehouse rent, and any obsolescence/write-off exposure on bespoke components) and share this explicitly with sales and design, so the trade-off between customisation-driven sales growth and inventory-carrying cost is visible at a decision-making level.
- **Review pricing for highly customised craft** so the true carrying cost of bespoke, low-volume component inventory is reflected in what those specific customers are charged, rather than absorbed across the whole product range.
- **Increase commonality across the bespoke component range** where design allows — a smaller set of components reused across more customisation combinations reduces the inventory needed to support the same variety a customer actually experiences.
- **Consider postponement or late-stage differentiation techniques** as the customisation programme scales, so bespoke features are finished as late as possible in the build process rather than held speculatively in advance as long-lead-time components.
- **Track lead-time and total cost of ownership alongside inventory turnover** going forward, since turnover alone shows that performance has moved, not why — the same limitation raised elsewhere in this module's inventory content.

Mark Allocation — Part (c) (worth 40% of the question)

Criterion	Marks	What earns the marks
Correct formula and 2019 calculation	6%	Turnover = COGS/Avg Inventory; $40/3 \approx 13.3x$ shown with working
Correct 2020 calculation	6%	$50/5 = 10.0x$ shown with working
Correctly identifies that performance has worsened	6%	States clearly that turnover fell despite COGS growth, and connects this to the reported margin erosion
Reasons for the decline	12%	Identifies the customisation/bespoke-component driver and links it to loss of commonality/pooling, referencing the warehouse space as evidence
Improvement measures	10%	At least three specific, relevant measures (e.g. cost visibility to sales/design, pricing review, commonality, postponement)
Total	40%	Sum of criteria above

Question 3 — total

100%